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Prediction of Nursing Need Proxies Using Vital Signs and Biomarkers Data: Application of Deep Learning Models

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ABSTRACT

Aim: To develop deep learning models to predict nursing need proxies among hospitalised patients and compare their predictive efficacy to that of a traditional regression model.

Design: This methodological study employed a cross-sectional secondary data analysis.

Methods: This study used de-identified electronic health records data from 20,855 adult patients aged 20 years or older, admitted to the general wards at a tertiary hospital. The models utilised patient information covering the preceding 2 days, comprising vital signs, biomarkers and demographic data. To create nursing need proxies, we identified the six highest-workload nursing tasks. We structured the collected data sequentially to facilitate processing via recurrent neural network (RNN) and long short-term memory (LSTM) algorithms. The STROBE checklist for cross-sectional studies was used for reporting.

Results: Both the RNN and LSTM predicted nursing need proxies more effectively than the traditional regression model. However, upon testing the models using a sample case dataset, we observed a notable reduction in prediction accuracy during periods marked by rapid change.

Conclusions: The RNN and LSTM, which enhanced predictive performance for nursing needs, were developed using iterative learning processes. The RNN and LSTM demonstrated predictive capabilities superior to the traditional multiple regression model for nursing need proxies.

Implications for the Profession: Applying these predictive models in clinical settings where medical care complexity and diversity are increasing could substantially mitigate the uncertainties inherent in decision-making processes.

Patient or Public Contribution: We used de-identified electronic health record data of 20,855 adult patients about vital signs, biomarkers and nursing activities.

Reporting Method: The authors state that they have adhered to relevant EQUATOR guidelines: STROBE statement for cross-sectional studies.

Impact: Despite widespread adoption of deep learning algorithms in various industries, their application in nursing administration for workload distribution and staffing adequacy remains limited. This study amalgamated deep learning technology to develop a predictive model to proactively forecast nursing need proxies. Our study demonstrates that both the RNN and LSTM models outperform a traditional regression model in predicting nursing need proxies. The proactive application of deep learning

methods for nursing need prediction could help facilitate timely detection of changes in patient nursing demands, enabling the effective and safe nursing services.

1 | Introduction

Nursing needs encompass the degree to which (quantity and frequency) patients require specific nursing interventions (Ko and Park 2020). Precisely evaluating nursing needs and equitably distributing nursing workloads are crucial to maintaining patient safety and generating favourable health outcomes. Given that patient safety and health outcomes serve as pivotal performance metrics for healthcare institutions (Park and Kim 2018), nursing needs comprise a crucial quality assessment parameter for healthcare systems.

2 | Background

Prior studies have sought to quantify nursing needs using methods known for their relatively robust reliability and validity. Researchers have employed several measures for this purpose, including the Therapeutic Intervention Scoring System-28 (TISS-28), Nine Equivalents of Nursing Manpower Use Score (NEMS) and Nursing Activities Score (NAS) (Ebrahimian et al. 2018; Miranda, de Rijk, and Schaufeli 1996; Urden and Roode 1997). Widely adopted in South Korea, the Korean Patient Classification System (KPCS) provides a framework for quantifying required nursing care and assessing its complexity based on the severity of patient conditions (Song et al. 2018). Nonetheless, concerns regarding the efficiency of these methods have surfaced, particularly related to the increased workload for nurses who must manually input the data (Perroca and Ek 2007).

To address this challenge, healthcare institutions have developed patient severity classification systems that leverage patient clinical data to automatically estimate nursing needs (Kim et al. 2022). Although researchers have reported successful automatic nursing need estimations on the basis of electronic health records (EHRs), differences in the EHR systems employed across healthcare organisations remain a persistent hurdle to implementation. Moreover, these classification outcomes often reflect patients' clinical statuses at specific points in time, which constrains their ability to dynamically capture changes in nursing needs as clinical conditions evolve. Thus, the development of a universally applicable nursing need prediction model that is adaptable to diverse EHR systems could have substantially enhanced utility and effectiveness.

The proactive application of deep learning methods for nursing need prediction could help facilitate timely detection of changes in patient nursing demands, enabling the efficient provision of personalised nursing services. Nursing need prediction informs data-driven decision-making regarding proactive staffing strategies and equitable workload distribution in healthcare settings. Moreover, it fosters preparedness for unforeseen events, potentially alleviating the psychological burden nurses experience when carrying out unpredictable care tasks. Despite widespread adoption of deep learning algorithms in various industries, their application in nursing administration for workload distribution and staffing adequacy remains limited. Researchers have extensively examined applications of deep learning for applications such as image-based diagnosis, clinical prognosis prediction and discharge date forecasting (Islam et al. 2019). However, direct applications of deep learning algorithms to nursing administration remain scarce.

3 | The Study

Given the inherent complexity and heterogeneity of nursing activities, directly and accurately assessing nursing needs is a significant challenge (Ross, Rogers, and King 2019). To address this, we use nursing need proxies as estimators of nursing demands, leveraging artificial intelligence (AI) technology to develop a prediction model for nursing need proxies among hospitalised patients. By assuming that patient clinical conditions inherently reflect nursing needs, a deep learning-based prediction model utilising EHR data encompassing vital signs and routine biomarkers may facilitate accurate nursing need estimation.

4 | Methods

4.1 | Design

This methodological research employed a cross-sectional secondary data analysis design to develop deep learning models to predict nursing need proxies and assess their predictive efficacy relative to a traditional regression model. The deep learning models utilised patient data spanning the preceding 2 days—that is, vital signs, biomarkers and demographic information, plus nursing need proxies. Figure 1 shows the conceptual framework guiding this investigation. The study design and findings were reported according to the Strengthening The Reporting of OBservational studies in Epidemiology (STROBE) checklist-guidelines for reporting cross-sectional studies (Data S1).

4.2 | Research Procedure

4.2.1 | Data Collection

We collected de-identified EHR data from 20,855 adult patients aged 20 years or older, admitted to the general wards at a tertiary hospital from November 2019 to October 2020. As tertiary hospitals provide acute medical services, our analysis focused on hospital stays up to 8 days, consistent with the average length of stay in healthcare institutions in South Korea (7.7 days). Consequently, we excluded hospital stays exceeding this threshold from the analysis (Health Insurance Review and Assessment Service 2023).

Summary

- What does this paper contribute to the wider global clinical community?
  - This study amalgamated deep learning technology to develop a predictive model to proactively forecast nursing need proxies and demonstrates that both the RNN and LSTM models outperform a traditional regression model.
  - The proactive application of deep learning methods for nursing need prediction could help facilitate timely detection of changes in patient nursing demands, enabling the effective and safe nursing services.

To develop nursing need proxies, we chose the six highest-workload nursing tasks as representative nursing activities from all nursing tasks extracted from the HER (An 2022). The selected nursing intervention items were (1) intravenous injection, (2) whole blood transfusion or packed red blood cell transfusion, (3) drainage tube management, (4) vital signs measurement, (5) admission nursing and (6) discharge nursing.

We selected vital signs and routine biomarkers reflecting the patient's clinical conditions on the basis of the Acute Physiology and Chronic Health Evaluation II (APACHE II) variables. Vital signs included mean arterial pressure (MAP), heart rate, respiratory rate and body temperature. For biomarkers, we used haematocrit, serum creatinine, serum sodium and serum potassium. To reflect overall infection status, we included C-reactive protein (CRP) because CRP reacts more sensitively to acute infection than white blood cells, making it a superior predictor of postoperative complications or clinical deterioration (Maheshwari et al. 2020). We measured oxygenation as percutaneous oxygen saturation (SpO_2), which is routinely measured in general wards. Among the APACHE II variables, we did not consider arterial pH and Glasgow Coma Scale for prediction variables, because the data were frequently unavailable from the general wards.

In addition to vital signs and biomarkers, we considered patient demographic data including age, primary specialty of care and number of hospital days in predicting nursing need proxies (Park 2005).

4.2.2 | Data Preprocessing

We treated outliers in the collected data as missing: for example, negative vital sign values or SpO_2 higher than 100%. We imputed all missing vital sign and biomarker values with the preceding values (Lipton et al. 2015). If the preceding values were absent, we replaced missing values with the values at the closest time steps to prevent a significant loss of data (Lipton et al. 2015). When blood tests were not conducted during hospitalisation, we replaced the missing values with the median values within the normal range (Lipton et al. 2015).

To measure the nursing need proxies, we first calculated the intensity multiplied by the number of occurrences for each of the six nursing tasks and then summed the six nursing task scores. The intensity of nursing tasks was estimated on the basis of the time and demands required to complete the task, and used that of intravenous injection as a reference (An 2022): nursing workload for intravenous injection (7.8), admission nursing (27.9), discharge nursing (20.7), whole blood transfusion or packed RBC transfusion (15.8), drainage tube management (3.4) and vital signs measurement (2.4). We determined nursing task frequency using the prescription count for intravenous injection, the number of prescriptions for whole blood transfusion or packed RBC transfusion and the drainage tube management number for different drainage

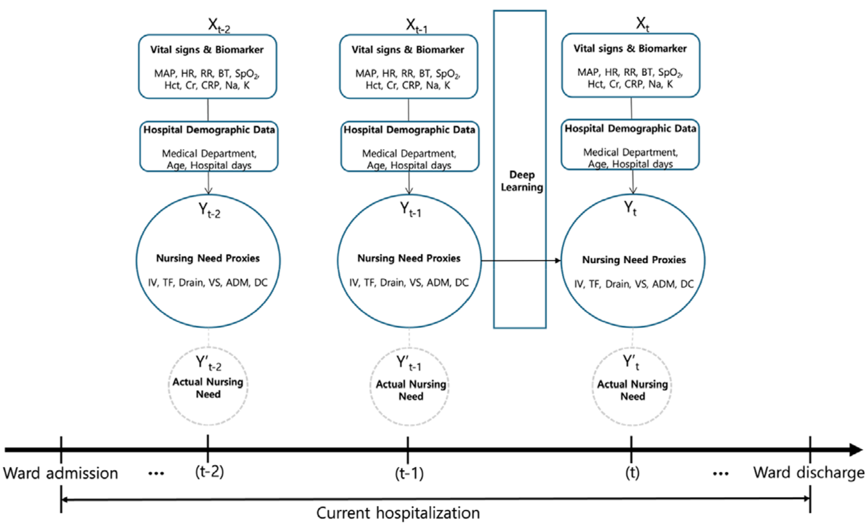

FIGURE 1 | Conceptual framework of the study. ADM, admission; BT, body temperature; Cr, serum creatinine; CRP, C-reactive protein; DC, discharge; Drain, number of drain tube; Hct, haematocrit; HR, heart rate; IV, number of intravenous injection orders; K, serum potassium; MAP, mean arterial pressure; Na, serum sodium; RR, respiratory rate; SpO2, saturation of percutaneous oxygen; *t*, prediction date; *t*-1, 1 day before the prediction; *t*-2, 2 days before the prediction; TF, whole blood and packed RBC transfusion; VS, number of vital signs recorded; *X*, data reflecting the patient's clinical status; *Y*′, actual nursing needs; *Y*, nursing need proxies. [Colour figure can be viewed at wileyonlinelibrary.com]

tube types. For the analytic purpose to assess changes in the distribution of nursing need proxies according to length of stay, we further categorised the proxies into four groups using quartiles: Group 1 for the lowest nursing need proxies and Group 4 for the highest nursing need proxies.

For vital signs such as MAP, heart rate, respiratory rate, body temperature and SpO_2, we used the minimum and maximum daily values. For biomarkers such as haematocrit, serum creatinine, CRP, serum sodium and serum potassium, we calculated average daily values.

To increase model performance, we normalised all vital signs, biomarkers and nursing need proxies (range 0–1) using Scikit-Learn's MinMaxScaler (Géron 2022). Figure S1 illustrates the data preparation and preprocessing processes.

4.2.3 | Configuration of Sequential Data

Recurrent neural network and LSTM algorithms are effectively utilised in processing medical data consisting of sequences of temporal measurements, for example, vital signs or biomarkers (Zhao et al. 2022). To facilitate processing via RNN and LSTM, we transformed the preprocessed data into sequential data, setting the sequence length and stride (Chollet 2021). Sequence length represents the number of time steps in the model input, whereas sequence stride denotes the time interval between the steps sliding across the entire dataset into the sequences during the partition process.

For short-term predictions using RNN or LSTM models, various sequence lengths of time-series data can be tested, ranging from a minimum of two steps to tens of steps (Hossain and Mahmood 2020; Huang, Huang, and Li 2019). Previously, Hossain and Mahmood (2020) focused on LSTM for short-term predictions, finding that while prediction performance was slightly degraded comparatively for sequence lengths set at two steps, the training time was significantly reduced. Maheshwari et al. (2020) employed LSTM with a sequence length of 48-time steps for data from the first 2 days of hospitalisation to predict the mortality rate for patients in the intensive care unit (ICU) (Maheshwari et al. 2020). We adopted this method to predict nursing need proxies on the basis of the information recorded over the initial 2 days of hospitalisation. However, setting time steps identical to the studies that targeted ICU patients proved challenging because general ward data is measured at longer intervals than ICU data. After considering the characteristics of the data in this study, we set 1 day as a single step and the sequence length as two. Meanwhile, we set the interval for generating sequences to 1 day, restructuring the dataset to reduce the cost of learning. The input data from the sequenced dataset consisted of vital signs, biomarkers, demographic information and nursing need proxies for each of the 2 days before the prediction date. We set nursing need proxies at the prediction date as target data, then normalised the sequenced dataset and conducted feature scaling for analysis. Next, we divided the dataset into a training dataset (70%), a validation dataset (20%) and a test dataset (10%), which we then utilised for model training, validation and evaluation, respectively (Yang et al. 2021).

4.2.4 | Data Analysis and Building Deep Learning Models

We conducted the frequency analysis and simulation test for the predictive model in the Google Colab environment using the Tesla V100 GPU (NVIDIA-SMI Version 525.105.17, Driver Version 525.105.17, CUDA Version 12.0), utilising the Statsmodels library (Version 0.14.0) to build the ordinary least squares multiple regression model. To develop the RNN and LSTM deep learning models, we used the TensorFlow (Version 2.14.0) and Keras (Version 2.14.0) libraries.

We determined the hyperparameters for each model by experimenting with various conditions to identify the best-performing values (Table S1). We used Adam, an efficient gradient descent algorithm combining the features of Momentum and RMSprop, with a learning rate of 0.001 and root mean squared error (RMSE) as the loss function (Kingma and Ba 2014). We set the epochs to 40 for RNN and 50 for LSTM, with batch sizes of 60 for the RNN and 40 for the LSTM during the training process.

4.2.5 | Model Evaluation

We evaluated the performance of the model using RMSE, which indicates the magnitude of prediction errors. We calculated RMSE value as the square root of the average of the squared differences between the predicted values and the actual values. A smaller RMSE value corresponds to more accurate model predictions. To identify the most suitable trendline that minimises the mismatch between predicted values and actual values, we compared the line of agreement between actual and predicted values with the trendlines of the models in terms of degree of alignment.

To assess the predictive performance of the models in a real case, we tested data regarding a 79-year-old surgical patient (Table S2).

4.3 | Ethical Consideration

We obtained approval for this study from the Institutional Review Board of Hallym University Medical Center (IRB protocol number: HALLYM 2021–12–021-001). During the study, we strictly adhered to the research ethics guidelines and conducted the study in accordance with the principles of the Declaration of Helsinki. We preserved all dataset anonymity.

5 | Results

5.1 | Characteristics of Collected Data

5.1.1 | Patients' Demographic Characteristics

The average age of the patients was 61.26 years (SD = 16.63), with a maximum age of 104 years. Approximately half of the patient population fell into the 50s (n = 5000, 23.98%) and 60s (n = 4432, 21.25%) age brackets. The study encompassed 14 distinct specialties of care: internal medicine, surgery, obstetrics and gynaecology, urology, otorhinolaryngology, neurology, rehabilitation medicine, ophthalmology, dentistry, dermatology, family medicine, emergency medicine, anaesthesiology and pain medicine, and nuclear medicine.

5.1.2 | Descriptive Statistical Analysis

In terms of vital signs, the MAP ranged from 19.33 to 223.67 mmHg, with average maximum and minimum values of 95.18 mmHg (SD = 12.18) and 83.13 mmHg (SD = 10.25), respectively (see Table 1). Heart rate ranged from 0 beats per minute to 263 beats per minute, with average maximum and minimum values of 82.78 beats per minute (SD = 13.27) and 72.06 beats per minute (SD = 10.39), respectively. Respiratory rate ranged from 0 to 60 breaths per minute, whereas body temperature ranged from 30.00°C to 41.70°C. Oxygen saturation (SpO2) ranged from 5% to 100%, with mean maximum and minimum values of 98.05% (SD = 1.60) and 97.12% (SD = 2.65), respectively.

Regarding biomarkers, the means of haematocrit and serum creatinine were 34.12 (SD = 5.45) and 0.99 mg/dL (SD = 1.26), respectively. C-reactive protein levels averaged at 23.80 (SD = 46.08); serum sodium and potassium levels were 138.70 (SD = 3.33) and 4.04 (SD = 0.46) in average, respectively.

5.1.3 | Distribution of Nursing Need Proxies

The distribution of nursing need proxies scores displayed significant skewness (skewness = 1.99), characterised by a pronounced right tail (Figure 2a) The range of scores varied notably, with a minimum of 2.40, a maximum of 664.40 and a mean of 59.57 (SD = 58.65). Patients fell predominantly into Groups 2 (47.85%) or 3 (37.62%) upon admission, with the proportions declining notably from the second days of their stays (Figure 2b).

5.2 | Multiple Linear Regression Models

The multiple linear regression model predicted nursing need proxies for the day after the preceding 2 days at a statistically significant level ($F = 2010.656$, $p < 0.001$) and its explanatory power was approximately 54% ($R^2 = 0.542$, ${}_{adj}R^2 = 0.540$) (Table 2). The Durbin–Watson statistic was 1.805, close to 2, suggesting no issue with the independence of residuals assumption. Notably, shorter hospital stay and older age on the previous day (*t*-1 day) were significantly associated with higher nursing need proxies ($B = -0.019$, $p < 0.001$ for hospital day; $B = 0.009$, $p < 0.001$ for age). In addition, the model identified all vital signs as significant predictors of nursing need proxies. Among the biomarkers, higher haematocrit and CRP, and lower serum sodium and potassium were significantly associated with higher nursing proxies, whereas the results for serum creatinine were not significant.

In the data from 2 days before the day of prediction (*t*-2 day), we found both length of stay and age to significantly influence nursing need proxies ($B = 0.018$, $p < 0.001$ for hospital day; $B = 0.009$, $p < 0.001$ for age). Regarding vital signs data, we found all variables except the maximum MAP, maximum heart rate and minimum SpO_2 values to be significantly related with nursing need proxies. Among the biomarkers, the regression coefficients of haematocrit and CRP were statistically significant.

TABLE 1 | Descriptive statistics for vital signs and biomarkers.

		Min	Max	Mean	Standard deviation
Vital signs	Max MAP	33.33	223.67	95.18	12.18
	Min MAP	19.33	163.33	83.13	10.25
	Max HR	7.00	263.00	82.78	13.27
	Min HR	0.00	235.00	72.06	10.39
	Max RR	0.00	60.00	19.72	2.32
	Min RR	0.00	46.00	18.69	1.74
	Max BT	33.60	41.70	37.08	0.52
	Min BT	30.00	40.40	36.59	0.38
	Max SpO_2	60.00	100.00	98.05	1.60
	Min SpO_2	5.00	100.00	97.12	2.65
Biomarkers	Mean Hct	10.60	56.60	34.12	5.45
	Mean Cr	0.16	17.61	0.99	1.26
	Mean CRP	0.50	301.00	23.80	46.08
	Mean Na	109.50	166.00	138.70	3.33
	Mean K	1.20	8.24	4.04	0.46

Abbreviations: BT, body temperature; Cr, serum creatinine; CRP, C-reactive protein; Hct, haematocrit; HR, heart rate; K, serum potassium; MAP, mean arterial pressure; Na, serum sodium; RR, respiratory rate; SpO_2, saturation of percutaneous oxygen.

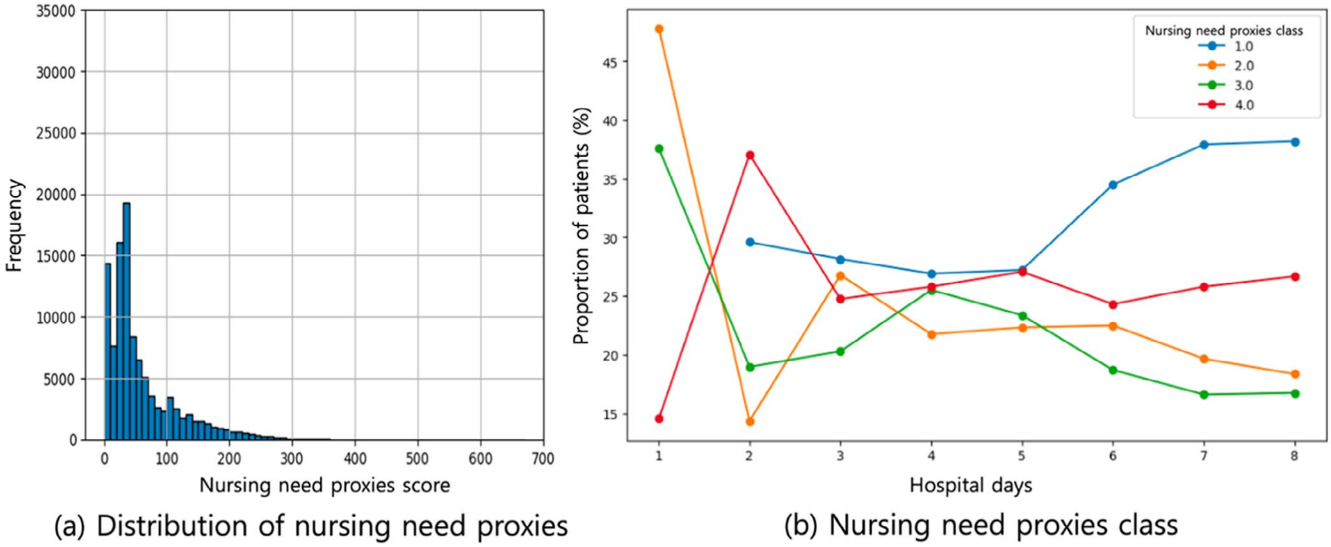

FIGURE 2 | Distribution of nursing need proxies. [Colour figure can be viewed at wileyonlinelibrary.com]

5.3 | Comparison of Predictive Performance Between the Multiple Regression Model and the Deep Learning Models

Figure 3 shows prediction errors for the number of training iterations in the RNN and LSTM models. For both RNN and LSTM, the prediction errors for the training and validation data decreased with repeated training.

Table 3 presents the prediction results for the multiple regression, RNN and LSTM models. The LSTM model exhibited the highest explanatory power at 61.2% ($R^2 = 0.612$, ${}_{\text{adj}}R^2 = 0.611$), followed by the RNN model at 57.5% ($R^2 = 0.575$, ${}_{\text{adj}}R^2 = 0.574$). The multiple regression model displayed the lowest explanatory power at 54.2% ($R^2 = 0.542$, ${}_{\text{adj}}R^2 = 0.540$). Evaluating the performances of all models, the LSTM model also demonstrated comparatively more accurate predictions on the test data (RMSE = 0.042), outperforming both the RNN model (RMSE = 0.044) and the multiple regression model (RMSE = 0.051).

We input the true and predicted values for all models in a scatter plot (Figure 4). The RNN and LSTM trendlines were closer to the line of agreement between actual and predicted values than multiple regression model trendline. Furthermore, the LSTM trendline was closer to the line of agreement between actual and predicted values than the RNN trendline.

5.4 | A Model Prediction Case

In the sample case data, vital signs were generally stable and within the normal range throughout the hospitalisation period (Table S3). CRP levels were elevated in the early stage of hospitalisation, reaching 108.79 mg/dL, but followed a decreasing trend beginning on the third day of hospitalisation. The nursing need proxies scores ranged from 40.50 to 318.80, with a mean of 110.61 (SD = 92.62). Nursing need proxies increased sharply on the third day of hospitalisation and then gradually decreased thereafter. Regarding the models' predictions regarding this case data, all three models poorly predicted the distinctively high nursing need proxies on the third day of hospitalisation. Their predicted scores for nursing need proxies were significantly lower than the actual values (Figure 5, Table S4). Subsequently, all three models predicted decreasing trends similar to the actual change during the remainder of the hospital stay.

6 | Discussion

In recent years, applications of AI technology have expanded exponentially across various domains, catalysing revolutionary transformations in society. Notably, AI-related research in nursing and health informatics has surged significantly during this period, as evidenced by recent studies (Chang, Jen, and Su 2022). In increasingly intricate and diverse clinical environments, AI—bolstered by big data analytics—has come to play a pivotal role in supporting evidence-based nursing care and thereby mitigating the uncertainties inherent in decision-making processes (Chang, Jen, and Su 2022). In this study, we amalgamated deep learning technology that implements AI to develop a predictive model to proactively forecast nursing need proxies. Our overarching objective was to facilitate data-driven nursing decision-making in hopes of optimising staffing levels and ensuring patients receive safe and effective care. Leveraging RNN and LSTM algorithmic models, we focused on forecasting nursing need proxies for the subsequent day by analysing trends spanning the preceding 2 days—vital signs, biomarkers, demographic data and existing nursing need proxies. Regarding nursing need proxies, we found that deep learning methodologies exhibited superior explanatory and predictive power than conventional multiple regression techniques. These findings highlight the distinct advantage of employing deep learning models, which can more effectively capture and model intricate patterns and interrelationships than traditional statistical methodologies.

Various prior studies have demonstrated the efficacy of the RNN and LSTM models. For instance, a study focusing on early-stage cancer occurrence prediction utilising time-series tumour biomarker data found that the LSTM model performed best among other machine-learning methodologies (Wu et al. 2022). Similarly,

TABLE 2 | Effect of features on nursing need proxies.

	Features	t-1 day			t-2 day		
		B	*t*	*p*	*B*	*t*	*p*
Patient	Hospital day	−0.019	−10.208***	< 0.001	0.018	9.655***	< 0.001
Demographics	Age	0.009	16.723***	< 0.001	0.009	16.723***	< 0.001
Vital signs	Max MAP	0.029	4.801***	< 0.001	−0.005	−0.888	0.374
	Min MAP	−0.039	−8.684***	< 0.001	0.014	3.412***	< 0.001
	Max HR	0.048	6.450***	< 0.001	0.004	0.558	0.577
	Min HR	−0.054	−9.986***	< 0.001	0.031	6.308***	< 0.001
	Max RR	−0.041	−4.789***	< 0.001	0.076	10.543***	< 0.001
	Min RR	0.111	17.414***	< 0.001	−0.074	−13.103***	< 0.001
	Max BT	0.672	24.882**	0.003	−0.192	−7.577***	< 0.001
	Min BT	−0.104	−3.474***	< 0.001	−0.173	−6.122***	< 0.001
	Max SpO_2	0.063	2.960***	< 0.001	−0.092	−4.362***	< 0.001
	Min SpO_2	0.039	3.341***	< 0.001	0.003	0.240	0.810
Biomarkers	Mean Hct	0.031	3.521***	< 0.001	−0.043	−4.901***	< 0.001
	Mean CRP	0.047	10.466***	< 0.001	0.024	5.576***	< 0.001
	Mean Cr	−0.005	−0.243	0.808	0.010	0.479	0.632
	Mean Na	−0.096	−3.614***	< 0.001	0.031	1.186	0.236
	Mean K	−0.049	−5.233***	< 0.001	0.004	0.399	0.690
Speciality	Family medicine	0.063	1.043	0.297	−0.066	−1.095	0.274
	Anaesthesiology medicine	0.065	4.965***	< 0.001	0.065	4.965***	< 0.001
	Urology	0.059	7.081***	< 0.001	0.059	7.081***	< 0.001
	Obstetric gynaecology	0.067	7.963***	< 0.001	0.067	7.963***	< 0.001
	Neurology	0.063	7.561***	< 0.001	0.063	7.561***	< 0.001
	Ophthalmology	0.060	7.127***	< 0.001	0.060	7.127***	< 0.001
	Emergency medicine	0.056	4.271***	< 0.001	0.056	4.271***	< 0.001
	Otolaryngology	0.062	7.401***	< 0.001	0.062	7.401***	< 0.001
	Rehabilitation medicine	0.064	7.383***	< 0.001	0.064	7.383***	< 0.001
	Dentistry	0.061	7.245***	< 0.001	0.061	7.245***	< 0.001
	Dermatology	0.074	8.487***	< 0.001	0.074	8.487***	< 0.001
	Nuclear medicine	−0.267	−4.956***	< 0.001	0.400	5.966***	< 0.001
	Medicine	−0.337	−8.763***	< 0.001	0.464	9.040***	< 0.001
	Surgery	−0.286	−9.428***	< 0.001	0.415	9.477***	< 0.001
Nursing need proxies		0.499	181.679***	< 0.001	0.068	24.729***	< 0.001

Note: $F = 2010.656$ ($p < 0.001$), $R^2 = 0.542$, $_{adj}R^2 = 0.540$, $D\text{-}W = 1.805$.
Abbreviations: BT, body temperature; Cr, serum creatinine; CRP, C-reactive protein; Hct, haematocrit; HR, heart rate; K, serum potassium; MAP, mean arterial pressure; Na, serum sodium; RR, respiratory rate; SpO_2, saturation of percutaneous oxygen.
**$p < 0.01$.
***$p < 0.001$.

in the realm of cardiovascular disease management, both simple the RNN and LSTM models exhibited superior predictive capabilities in forecasting medication adherence (Hsu, Warren, and Riddle 2022). Notably, the LSTM model demonstrated greater predictive accuracy than the RNN model. Consistent with prior findings, we found that the LSTM model had better predictive performance than the RNN model concerning nursing need proxies. This disparity in performance could stem from the inherent

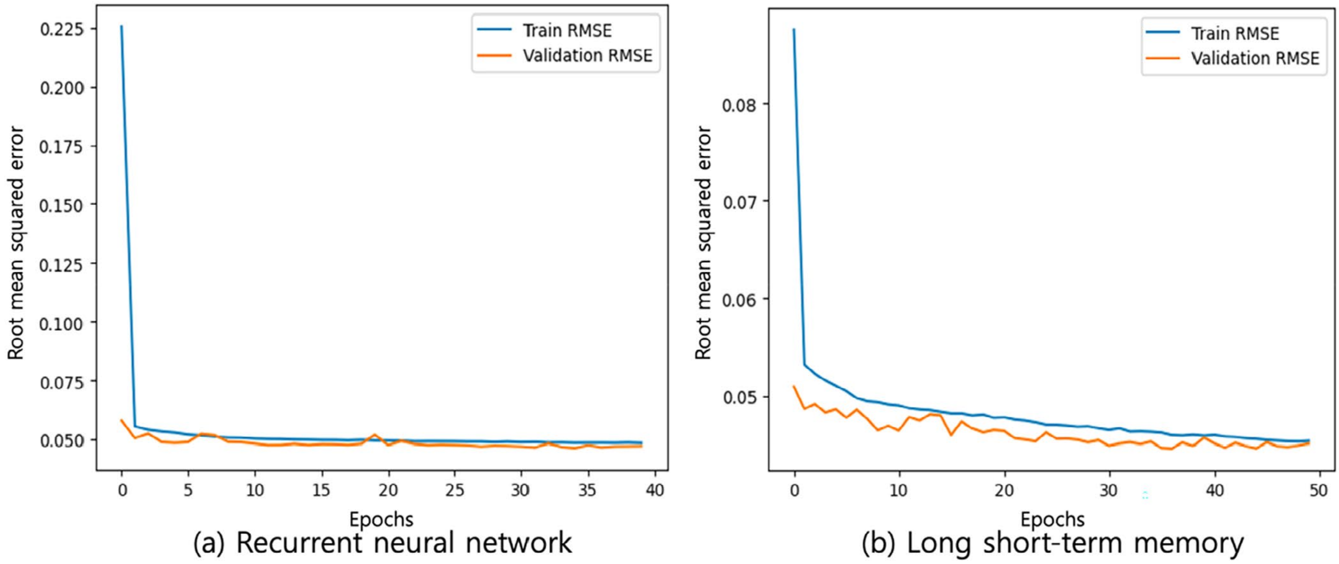

FIGURE 3 | Root mean squared error of recurrent neural network and long short-term memory models. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 3 | Comparison of model performance.

	R^2	$_{adj}R^2$	RMSE
Linear regression	0.542	0.540	0.051
Recurrent neural network	0.575	0.574	0.044
Long short-term memory	0.612	0.611	0.042

Abbreviation: RMSE, root mean squared error.

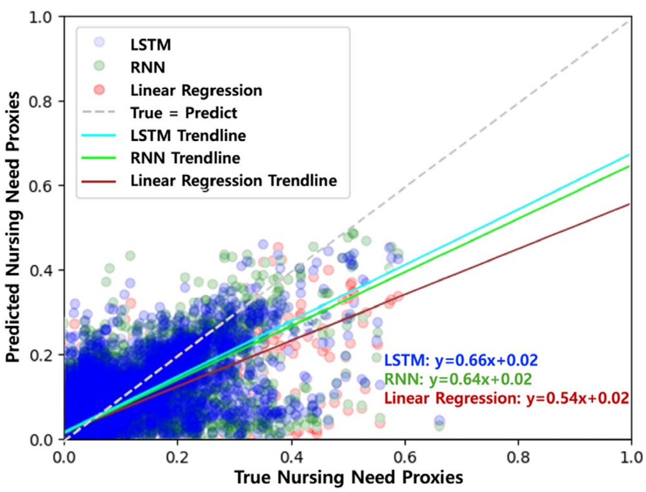

FIGURE 4 | Scatter plot of models. [Colour figure can be viewed at wileyonlinelibrary.com]

strengths of the LSTM model in addressing long-term dependency issues, which are generated by its distinctive cell state and gate mechanisms—capabilities absent in traditional RNN architectures (Rao, Rao, and Reddy 2022).

Notably, our research findings underscore the significant influence of patient age and care specialty on the predictive accuracy of nursing need proxies. This echoes previous research highlighting these two factors in nursing severity among hospitalised cancer patients (Park 2005). As patient condition severity levels tend to increase with age (Jin 2021), older patients generally require more complex treatments and intensive monitoring, contributing to an upward trend in nursing need proxies.

Additionally, we found that nursing need proxies tended to decrease as hospital stays lengthened. Patients in the early stages of hospitalisation are often in acute conditions and show frequent fluctuations in vital signs, requiring various nursing care for stabilisation (Loisa et al. 2022). In the case of vital signs, data closer to the prediction date (i.e., values on *t*-1 rather than *t*-2) were more likely to be significant for nursing need proxies. Presumably, this is a result of the dynamic nature of vital signs, characterised by frequent fluctuations that mirror the acute phase condition.

Although serum creatinine has traditionally served as a crucial variable in forecasting the severity of ICU patients, as evidenced by its inclusion in APACHE II, we did not find it to be significant in predicting nursing need proxies among patients in general wards. This finding is consistent with prior research (Patel et al. 2022) observing that the mortality rates of patients with acute kidney injuries were predicted by factors such as MAP, heart rate, respiratory rate and APACHE II scores, but not serum creatinine levels. These results suggest that although serum creatinine is included in APACHE II to reflect the chronic conditions of ICU-admitted patients, nursing need proxies are more sensitive to real-time changes in patient status, meaning they are less impacted by variables reflecting chronic conditions.

Notably, when we applied the prediction models to a sample case dataset, we observed a sharp rise in nursing need proxies on the second day of hospitalisation, followed by a subsequent decreasing trend. Furthermore, the model's prediction accuracy appeared to diminish during periods characterised by rapid changes. Considering the documented increases in postoperative nursing tasks (Cha 2023), the abrupt fluctuations in nursing need proxies may stem from the incorporation of surgery

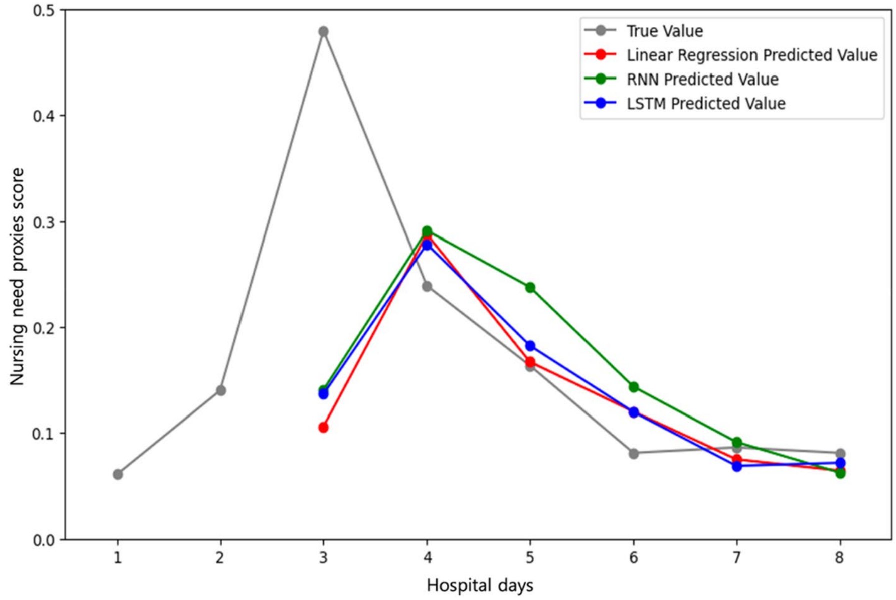

FIGURE 5 | Actual and predicted sample case values. [Colour figure can be viewed at wileyonlinelibrary.com]

schedules during the hospitalisation period. To enhance prediction precision and facilitate the interpretation of nursing need proxy alterations, future studies should consider including additional data, such as information pertaining to surgical necessity, primary diagnosis and underlying medical conditions.

Our deep learning-based predictive models are capable of learning nonlinear relationships and modelling complex patterns, making them suitable for forecasting the workload and complexity of nursing tasks that exhibit unstructured characteristics (Chollet 2021). The predictive results of these models can help reduce uncertainty in the decision-making process and facilitate data-driven nursing decisions. They can also be utilised to assist in decision-making for workforce allocation (National Health Insurance Service 2022), thereby enabling a balanced distribution of nurses' workloads to effectively ensure the quality of care (Tubbs-Cooley et al. 2019). By using the model developed in this study to flexibly allocate nursing resources in response to varying nursing needs, healthcare institutions have the potential to improve operational efficiency.

Although the models investigated in this study hold promise for informing staffing decisions and thereby helping to enhance nursing care quality, the limitations of this study warrant mention. Deep learning models demonstrate remarkable learning capabilities, particularly when applied to large, complex datasets. However, our constrained selection of independent variables may have generated data that lacked sufficient complexity, potentially limiting the models' learning capacities and, consequently, their predictive performance. Moreover, although LSTM models are better suited for processing longer sequences than traditional RNNs, our study attempted predictions on the basis of data from only 2 days. As a result, the performance enhancement stemming from structural disparities between models might have been marginal. To bolster model efficacy, future studies should endeavour to enrich the complexity of their datasets by incorporating a broader array of features, including primary diagnosis and surgical data, and extending prediction horizons to encompass sequences of adequate length. In addition, our interpretations of the RNN and LSTM models were hindered by the absence of information regarding the magnitude of the factors' impacts on the dependent variable (Castelvecchi 2016). To strengthen the credibility of decision-making processes within nursing administration, it is imperative to leverage advanced methodologies such as Shapley Additive Explanations and Attention Mechanism. These techniques can enhance the interpretability of the models and facilitate the exploration of additional influential variables (Liu, Bi, and Gui 2024).

7 | Conclusions

Our study developed RNN and LSTM models that improve the predictive performance for nursing needs proxies through iterative learning. Both the RNN and LSTM models demonstrated superior performance compared with traditional regression models in predicting nursing need proxies. Despite the inherent limitations of deep learning models, including their relatively low explanatory power and black-box nature, we innovatively applied RNN and LSTM architectures to forecast nursing need proxies and validated their efficacy. Future studies should examine the efficacy of diverse deep learning models in predicting nursing need proxies and focus on optimising hyperparameters to enhance predictive accuracy.

Author Contributions

Yunmi Baek: conceptualisation, formal analysis, methodology, investigation, visualisation, writing – original draft, writing – review and editing, project administration. **Kihye Han:** conceptualisation, investigation, writing – original draft, writing review and editing, project administration, supervision, funding acquisition. **Eunjoo Jeon:** conceptualisation, methodology, investigation, writing – review and editing. **Hae Young Yoo:** conceptualisation, methodology, investigation, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated and/or analysed during the current study are not publicly available due to confidentiality of the hospital patient data.

Statistics

The authors have checked to make sure that our submission conforms as applicable to the Journal's statistical guidelines. (b) There is a statistician on the author team (Dr. Eunjoo Jeon).

References

An, M. J. 2022. "Prediction Model of Nursing Workload Using Nursing Activities and Relative Values." Doctoral Dissertation, Yonsei Graduate School. https://ir.ymlib.yonsei.ac.kr/handle/22282913/189805.

Castelvecchi, D. 2016. "Can We Open the Black Box of AI?" *Nature News* 538, no. 7623: 20–23. https://www.nature.com/news/can-we-open-the-black-box-of-ai-1.20731.

Cha, H. J. 2023. "Comparison of Nursing Intensity According to Medical Units in a Tertiary Hospital." Master, Seoul National University. https://hdl.handle.net/10371/196201.

Chang, C. Y., H. J. Jen, and W. S. Su. 2022. "Trends in Artificial Intelligence in Nursing: Impacts on Nursing Management." *Journal of Nursing Management* 30, no. 8: 3644–3653. https://doi.org/10.1111/jonm.13770.

Chollet, F. 2021. *Deep Learning With Python*. New York, NY: Simon and Schuster.

Ebrahimian, A., A. Fakhr-Movahedi, R. Ghorbani, and H. Ghasemian-Nik. 2018. "Investigating the Application of a Nine Equivalents of Nursing Aanpower Use Score to Identify Patients at the End Stages of Life." *Middle East Journal of Rehabilitation and Health* 5, no. 3: 1–6. https://doi.org/10.5812/mejrh.65153.

Géron, A. 2022. *Hands-On Machine Learning With Scikit-Learn, Keras, and TensorFlow*. Sebastopol, CA: O'Reilly Media, Inc.

Health Insurance Review and Assessment Service. 2023. "Key Indicators by Type of Hospitalization/Outpatient." https://opendata.hira.or.kr/op/opc/olapHthInsRvStatInfoTab2.do?docNo=03-002.

Hossain, M. S., and H. Mahmood. 2020. "Short-Term Load Forecasting Using an LSTM Neural Network." 2020 IEEE Power and Energy Conference at Illinois (PECI), 1–6.

Hsu, W., J. R. Warren, and P. J. Riddle. 2022. "Medication Adherence Prediction Through Temporal Modelling in Cardiovascular Disease Management." *BMC Medical Informatics and Decision Making* 22, no. 1: 1–21. https://doi.org/10.1186/s12911-022-02052-9.

Huang, C.-G., H.-Z. Huang, and Y.-F. Li. 2019. "A Bidirectional LSTM Prognostics Method Under Multiple Operational Conditions." *IEEE Transactions on Industrial Electronics* 66, no. 11: 8792–8802. https://ieeexplore.ieee.org/abstract/document/8611256.

Islam, M. S., H. M. Umran, S. M. Umran, and M. Karim. 2019. "Intelligent Healthcare Platform: Cardiovascular Disease Risk Factors Prediction Using Attention Module Based LSTM." 2019 2nd International Conference on Artificial Intelligence and Big Data (ICAIBD) (pp. 167–175). IEEE. https://ieeexplore.ieee.org/abstract/document/8836998.

Jin, H. K. 2021. "Nursing Activities and Nursing Hours Provided to COVID-19 Inpatients in a Government-Designated Isolation Unit." Masters's Thesis, Seoul National University. http://dcollection.snu.ac.kr:8080/ezpdfdrm/dCollection.jsp?sItemId=000000164524.

Kim, J., T. Kang, H.-J. Seo, et al. 2022. "Measuring Patient Acuity and Nursing Care Needs in South Korea: Application of a New Patient Classification System." *BMC Nursing* 21, no. 1: 332. https://doi.org/10.1186/s12912-022-01109-4.

Kingma, D. P., and J. Ba. 2014. "Adam: A Method for Stochastic Optimization." https://doi.org/10.48550/arXiv.1412.6980.Preprint *ArXiv*:1412.6980.

Ko, Y., and B. Park. 2020. "Calculation of Optimum Number of Nurses Based on Nursing Intensity of Intensive Care Units." *Korea Journal of Hospital Management* 25, no. 3: 14–28. https://koreascience.kr/article/JAKO202029062617511.page.

Lipton, Z. C., D. C. Kale, C. Elkan, and R. Wetzel. 2015. "Learning to Diagnose With LSTM Recurrent Neural Networks." https://doi.org/10.48550/arXiv.1511.03677. Preprint ArXiv:1511.03677.

Liu, L., B. Bi, and M. Gui. 2024. "Predictive Model and Risk Analysis for Peripheral Vascular Disease in Type 2 Diabetes Mellitus Patients Using Machine Learning and Shapley Additive Explanation." *Frontiers in Endocrinology* 15: 1320335. https://doi.org/10.3389/fendo.2024.1320335.

Loisa, E., A. Kallonen, S. Hoppu, and J. Tirkkonen. 2022. "Trends in the National Early Warning Score Are Associated With Subsequent Mortality–a Prospective Three-Centre Observational Study With 11,331 General Ward Patients." *Resuscitation Plus* 10: 100251. https://doi.org/10.1016/j.resplu.2022.100251.

Maheshwari, S., A. Agarwal, A. Shukla, and R. Tiwari. 2020. "A Comprehensive Evaluation for the Prediction of Mortality in Intensive Care Units With LSTM Networks: Patients With Cardiovascular Disease." *Biomedical Engineering/Biomedizinische Technik* 65, no. 4: 435–446. https://doi.org/10.1515/bmt-2018-0206.

Miranda, D. R., A. de Rijk, and W. Schaufeli. 1996. "Simplified Therapeutic Intervention Scoring System: The TISS-28 Items-Results From a Multicenter Study." *Critical Care Medicine* 24, no. 1: 64–73. https://journals.lww.com/ccmjournal/abstract/1996/01000/simplified_therapeutic_intervention_scoring.12.aspx.

National Health Insurance Service. 2022. "Guidelines for the Integrated Nursing Care Service Project." https://www.nhis.or.kr/nhis/together/wbhaea01000m01.do?mode=view&articleNo=10815472.

Park, M. Y., and E. Y. Kim. 2018. "Perception of Importance of Patient Safety Management, Patient Safety Culture and Safety Performance in Hospital Managerial Performance of Hospital Nurses." *Journal of Korean Academy of Nursing Administration* 24, no. 1: 40–50. https://doi.org/10.11111/jkana.2018.24.1.40.

Park, S. A. 2005. "An Analysis of Nursing Needs for Hospitalized Cancer Patients: Using Data Mining Techniques." *Asian Oncology Nursing* 5, no. 1: 3–10. https://koreascience.kr/article/JAKO200533155710178.page.

Patel, P., S. Gupta, H. Patel, and M. A. Bashar. 2022. "Assessment of APACHE II Score to Predict ICU Outcomes of Patients With AKI: A Single-Center Experience From Haryana, North India." *Indian Journal of Critical Care Medicine* 26, no. 3: 276–281.

Perroca, M. G., and A. C. Ek. 2007. "Utilization of Patient Classification Systems in Swedish Hospitals and the Degree of Satisfaction Among Nursing Staff." *Journal of Nursing Management* 15, no. 5: 472–480. https://doi.org/10.1111/j.1365-2834.2007.00732.x.

Rao, S. U. M., K. V. Rao, and P. Reddy. 2022. "Medical Big Data Analysis Using LSTM Based Co-Learning Model With Whale Optimization Approach." *International Journal of Intelligent Engineering & Systems* 15, no. 4: 627–636. https://inass.org/wp-content/uploads/2022/05/2022083156-2.pdf.

Ross, C., C. Rogers, and C. King. 2019. "Safety Culture and an Invisible Nursing Workload." *Collegian* 26, no. 1: 1–7. https://doi.org/10.1016/j.colegn.2018.02.002.

Song, K. J., W. H. Choi, E. H. Choi, et al. 2018. "Study for Revision of the Korean Patient Classification System." *Journal of Korean Clinical*

Nursing Research 24, no. 1: 113–126. https://koreascience.kr/article/JAKO201813164519533.pdf.

Tubbs-Cooley, H. L., C. A. Mara, A. C. Carle, B. A. Mark, and R. H. Pickler. 2019. "Association of Nurse Workload With Missed Nursing Care in the Neonatal Intensive Care Unit. JAMA." *Pediatrics* 173, no. 1: 44–51. https://doi.org/10.1001/jamapediatrics.2018.3619.

Urden, L. D., and J. L. Roode. 1997. "Work Sampling: A Decision-Making Tool for Determining Resources and Work Redesign." *Journal of Nursing Administration* 27, no. 9: 34–41. https://journals.lww.com/jonajournal/abstract/1997/09000/work_sampling___a_decision_making_tool_for.9.aspx.

Wu, X., H.-Y. Wang, P. Shi, et al. 2022. "Long Short-Term Memory Model–a Deep Learning Approach for Medical Data With Irregularity in Cancer Predication With Tumor Markers." *Computers in Biology and Medicine* 144: 105362. https://doi.org/10.1016/j.compbiomed.2022.105362.

Yang, T. Y., P.-Y. Kuo, Y. Huang, et al. 2021. "Deep-Learning Approach to Predict Survival Outcomes Using Wearable Actigraphy Device Among End-Stage Cancer Patients." *Frontiers in Public Health* 9: 1–9. https://doi.org/10.3389/fpubh.2021.730150.

Zhao, F., X. Yu, J. Zhang, X. Li, and R. Li. 2022. "A Disease Progression Prediction Model Based on EHR Data. 2022 IEEE 24th Int Conf on High Performance Computing & Communications; 8th Int Conf on Data Science & Systems; 20th Int Conf on Smart City; 8th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys)." https://ieeexplore.ieee.org/abstract/document/10074885.

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